

- 5 (a) when the load is removed then the wire / body object does not return to its original shape / length B1 [1]
- (b) (i) stress = force / area C1
 $F = 220 \times 10^6 \times 1.54 \times 10^{-6} = 340 \text{ (338.8) N}$ A1 [2]
- (ii) $E = (F \times l) / (A \times e)$ C1
 $e = (90 \times 10^6) \times 1.75 / (1.2 \times 10^{11}) = 1.31 \times 10^{-3} \text{ m}$ A1 [2]
- (c) the stress is no longer proportional to the extension B1 [1]

Page 4	Mark Scheme	Syllabus	Paper
	GCE AS/A LEVEL – October/November 2012	9702	21

6 (a) 92 protons in the nucleus and 92 electrons around nucleus

P1